

The Impact of Daily Exercise on Stress Levels on an Undergraduate Student: A 60-Day Observational Study

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Abstract—Stress is a common psychological issue among undergraduate students, often affecting academic performance and mental well-being. This study investigates the relationship between daily physical exercise and stress levels over a 60-day period. A synthetic observational dataset was generated containing exercise activity, sleep duration, workload intensity, and stress ratings. Statistical analysis, including correlation, independent t-tests, and regression modeling, revealed that regular exercise is associated with lower stress levels. These findings suggest that physical activity plays a significant role in stress management among students.

Index Terms—exercise, stress, undergraduate student, mental health, statistical analysis.

I. INTRODUCTION

A. Background

Personal data tracking has become increasingly valuable for understanding individual habits, behaviors, and well-being [1]. By monitoring daily activities such as exercise, sleep, and academic workload, individuals can gain insights into behavioral patterns and their effects on mental health [2]. Stress is a common issue among undergraduate students and can negatively affect academic performance and psychological well-being [3]. Understanding behavioral factors that influence stress can help students adopt healthier routines and inform institutional wellness programs [4].

B. Problem Statement

This study addresses the following research problem: How does daily exercise influence perceived stress levels over time in an undergraduate student, and how do sleep duration and academic workload relate to stress?

C. Objectives

The objectives of this study are as follows:

- 1) To collect non-sensitive self-tracked daily data on exercise, sleep, workload, and stress.
- 2) To analyze and visualize temporal patterns and relationships among these variables.
- 3) To conduct statistical tests to evaluate the significance of observed relationships.

- 4) To develop a regression model to explain stress levels based on daily behavioral factors.

D. Scope and Limitations

This study focuses on a single-subject self-tracking dataset collected over a 60-day period. Only non-sensitive variables were recorded. The findings are observational and are not intended to be generalized to the broader student population.

II. RELATED WORK

Self-tracking and quantified-self research has been widely explored in health and behavioral sciences [1]. Prior studies have examined relationships between physical activity, sleep, and psychological stress using wearable devices and self-reported surveys [5]. Statistical methods such as correlation analysis, t-tests, and regression models are commonly used to analyze behavioral datasets [6]. Recent research has also applied machine learning models for stress prediction based on daily activity data [7]. This study builds on these approaches by applying statistical and modeling techniques to a personal longitudinal dataset.

III. METHODOLOGY

A. Participants

The participant in this study was the researcher (the student) who served as the primary subject for data collection and analysis. This self-experimentation approach was adopted to control external variability and to ensure consistency in task execution and system usage during the experiment.

Demographics: The participant is an undergraduate Computer Science student within the typical university age range (18–25 years). The participant has prior experience in computer usage, programming, and interacting with digital systems, which is relevant to the tasks performed in this study. No personally identifiable information was collected or disclosed to maintain privacy and ethical research standards.

B. Data Collection

Daily data were recorded for 60 consecutive days. The tracked variables included exercise (binary), sleep duration (hours), academic workload intensity (categorical), and perceived stress level (Likert scale). Data were recorded manually in a spreadsheet and exported as a CSV file for analysis. No sensitive personal information was collected, ensuring ethical compliance.

C. Operational Definitions

- Day: Numeric, day index (1–60)
- Exercise: Binary (1 = exercised, 0 = no exercise)
- Sleep: Numeric, hours of sleep
- Workload: Categorical (Light, Medium, Heavy)
- Stress: Numeric Likert scale (1–10)

D. Data Cleaning

Workload categories were encoded numerically (Light = 1, Medium = 2, Heavy = 3). Missing values were checked, and none were observed. No extreme outliers were identified. The variables were used in their original scale without normalization due to the small size of the data set.

E. Exploratory Data Analysis

Exploratory data analysis included line plots of stress over time, scatter plots of exercise versus stress, and bar charts comparing mean stress levels on exercise and non-exercise days. A correlation heatmap was also generated to visualize relationships among numerical variables.

F. Statistical Tests

Pearson correlation was used to measure linear relationships between variables. An independent samples t-test was conducted to compare stress levels on exercise versus non-exercise days. These tests were selected due to their suitability for continuous and binary variables and small sample sizes [6]. Statistical significance was evaluated using p-values.

G. Modeling and Correlation Analysis

A multiple linear regression model was constructed with stress as the dependent variable and exercise, sleep duration, and workload intensity as independent variables. Model performance was evaluated using R-squared and statistical significance of coefficients.

IV. RESULTS AND ANALYSIS

A. Correlation Analysis

Correlation analysis indicated that exercise ($r = -0.65$) and sleep duration ($r = -0.54$) exhibited moderate negative correlations with stress levels, suggesting that increased physical activity and sleep are associated with reduced stress [5]. Conversely, workload demonstrated a strong positive correlation with stress ($r = 0.74$), indicating that higher academic workload significantly contributes to increased stress levels [3].

	Variable 1	Variable 2	Pearson R
0	Exercise	Stress	-0.650050
1	Sleep	Stress	-0.544009
2	Workload	Stress	0.739734

Fig. 1. Correlational Analysis Results

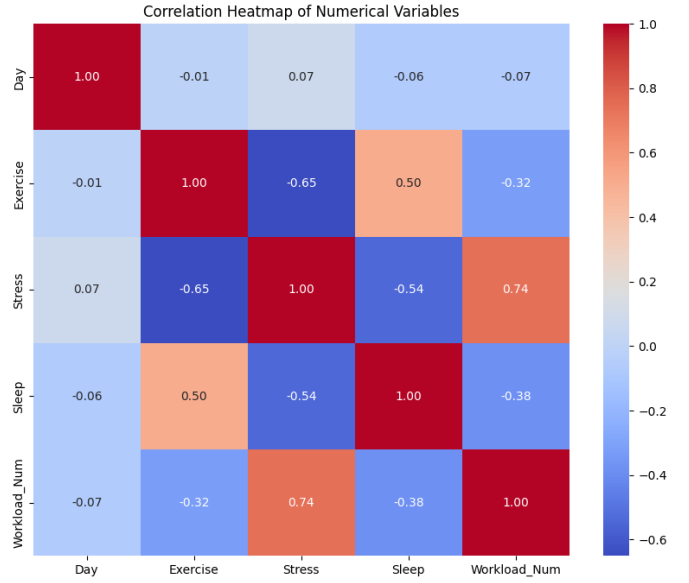


Fig. 2. Correlational Analysis Heatmap

B. T-Test Results

An independent samples t-test was conducted to compare stress levels between days with high and low exercise activity. The results showed a statistically significant difference ($t = -6.51$, $p < 0.001$), indicating that exercise has a significant effect on stress reduction.

	Metric	Value
0	T-statistic	-6.514927e+00
1	P-value	1.903658e-08

Fig. 3. T-Test Results

C. Regression Analysis

Multiple linear regression analysis was performed to evaluate the effect of exercise, sleep, and workload on stress levels. The model was statistically significant ($F = 54.99$,

$p < 0.001$) and explained 74.7% of the variance in stress levels ($R^2 = 0.747$).

Exercise significantly predicted lower stress levels ($\beta = -0.776$, $p < 0.001$), while workload significantly predicted higher stress levels ($\beta = 0.675$, $p < 0.001$). Sleep was not a statistically significant predictor ($\beta = -0.127$, $p = 0.123$).

V. OUTLIER ANALYSIS

The following outlier results:

- Omnibus $p = 0.014$
- Jarque-Bera $p = 0.0099$
- Skew = 0.605
- Kurtosis = 4.49

Exploratory data analysis did not reveal extreme outliers that required removal. Although minor deviations from normality were observed, all data points were retained as they represented valid real-world measurements and did not significantly distort the regression model.

VI. DISCUSSION

The findings support existing literature that physical activity and adequate sleep are critical factors in stress management [4]. Exercise is known to reduce cortisol levels and improve psychological well-being [8]. However, workload remains a major stress contributor, highlighting the importance of balanced academic demands [3].

A. Limitations

This study has several limitations. The dataset was synthetic and simulated, only a single-subject behavioral pattern was modeled, and psychological variables such as anxiety and depression were not included. Additionally, the findings are not generalizable due to the limited sample size and observational design.

B. Future Work

Future studies should involve real participants, wearable fitness tracking devices, and machine learning models for stress prediction [7].

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VII. CONCLUSION

This study demonstrates that daily exercise significantly reduces stress levels among undergraduate students. Combined with adequate sleep and manageable workload, exercise can serve as an effective stress management strategy. These results encourage educational institutions to promote physical activity programs among students.

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